

Supplementary Material 1

GEO Profiles of Creatine-related genes

We started a search within the the [GeoProfiles Database]

(<https://www.ncbi.nlm.nih.gov/geoprofiles>) using each gene symbol as main search term, using the following criteria:

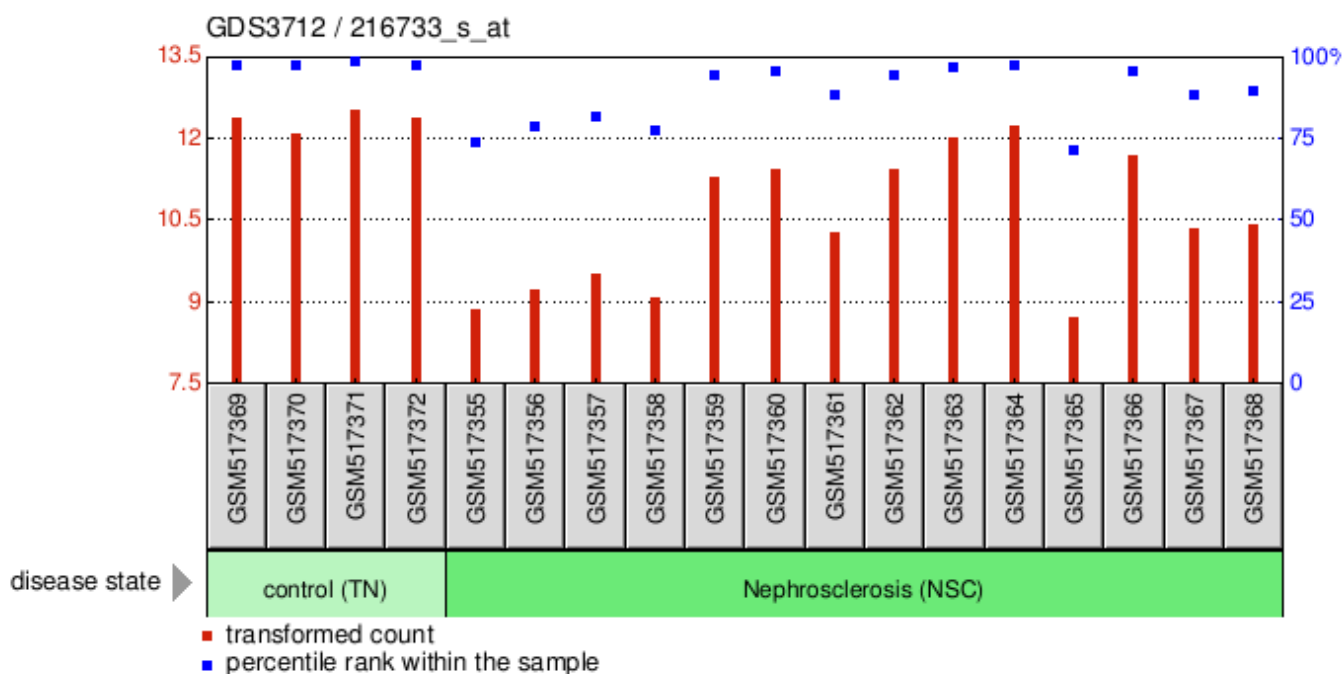
- Organism: *Homo sapiens*.
- Differential expression: Up/down genes.
- DataSet keyword: Kidney and/or Renal.
- Exclusion criteria: expression undetected, experimental design not comparing disease/metabolic/chemical treatment conditions or with mutants cell lines. We then merged the results, excluded any "renal" name variations (i.e. adrenal), and counted the overlapped profiles.

The results per gene will be reported bellow.

GATM

Using the aforementioned criteria, we obtained 16 distinct profiles for the GATM gene (9 for kidney, 10 for renal, 3 registers were present in both - marked with an *).

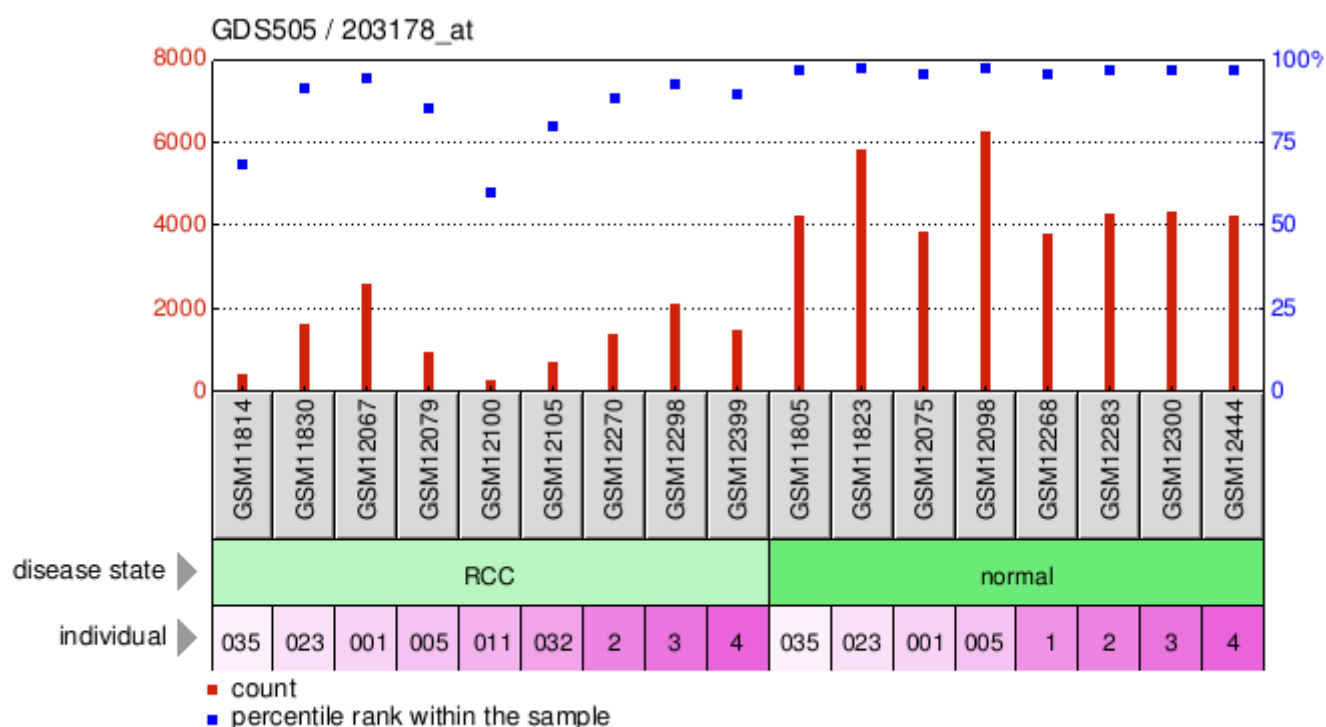
1. Nephrosclerosis: glomeruli (GDS3712 / 216733_s_at)*



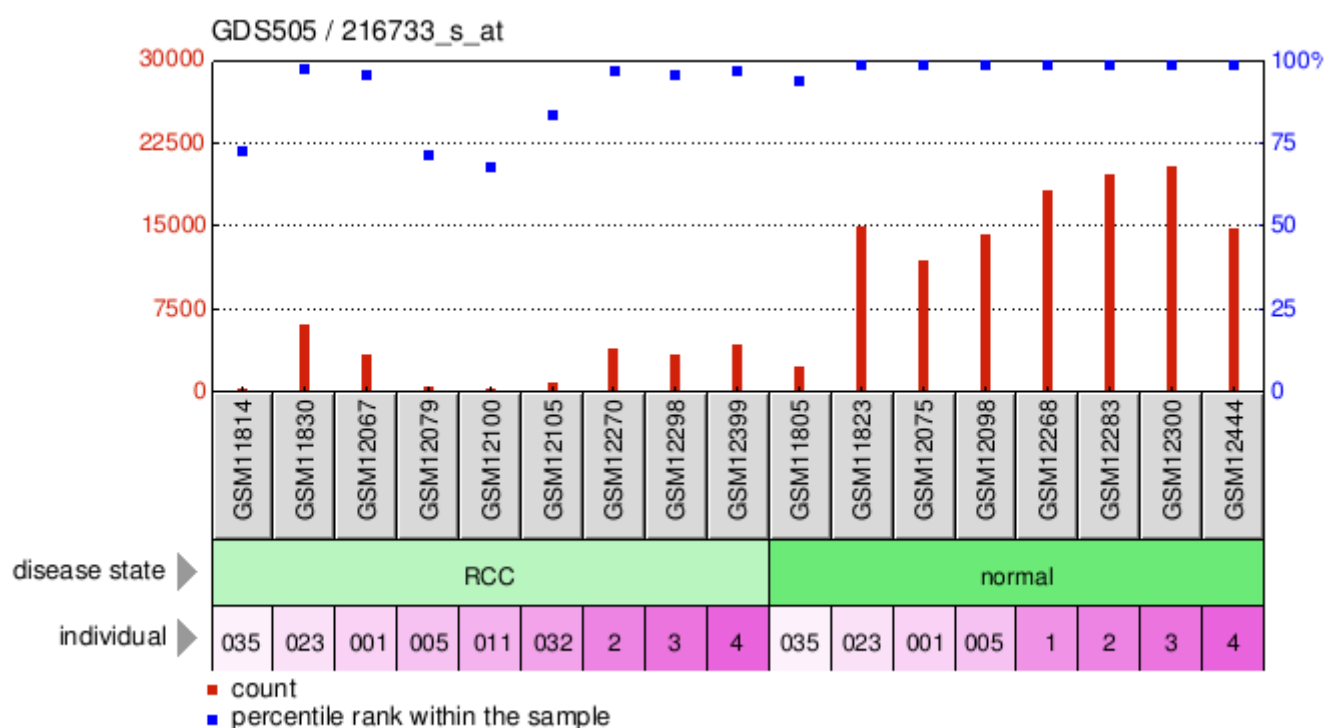
From the GEO-generated graph we can perceive a significative downregulation of the gene GATM from Nephrosclerosis samples, when compared to the control ones.

2. Renal clear cell carcinoma (HG-U133A) - GDS505*

Two profiles for the gene GATM: 2.1. GDS505 / 203178_at*



2.2. GDS505 / 216733_s_at*

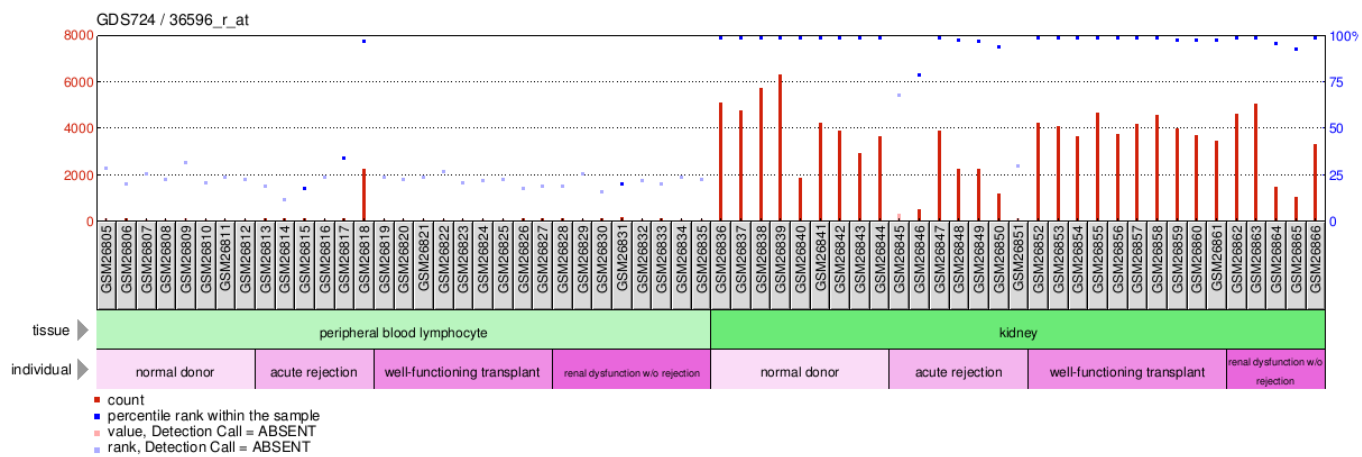


Both transcripts for GATM were downregulated when compared renal clear cell carcinomas (RCCs) versus control samples.

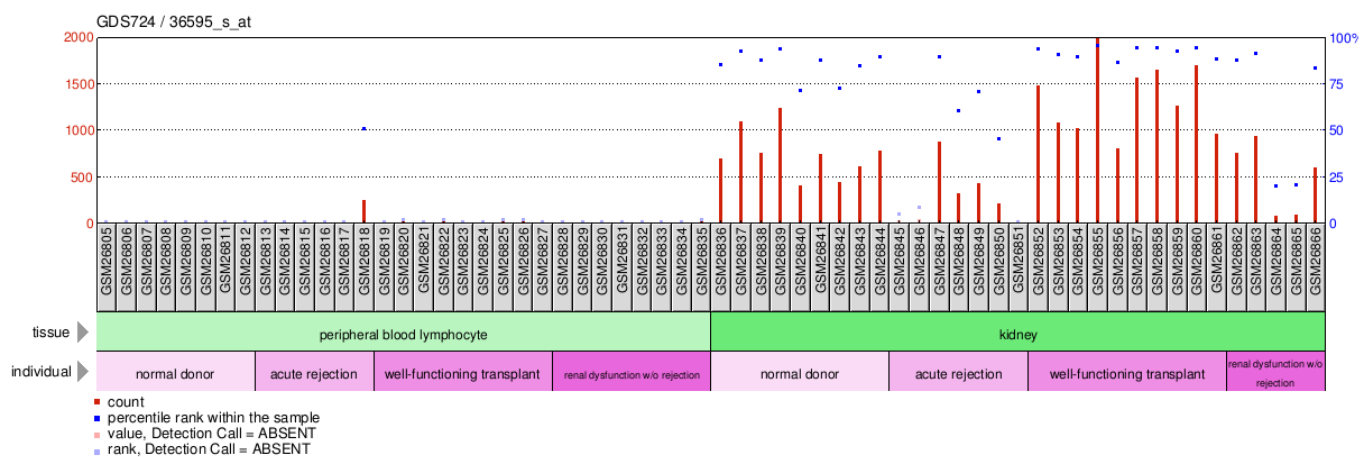
3. Kidney transplant rejection expression profiling (GDS724 / 36596_r_at)

Two profiles for the gene GATM:

3.1. GDS724 / 36596_r_at



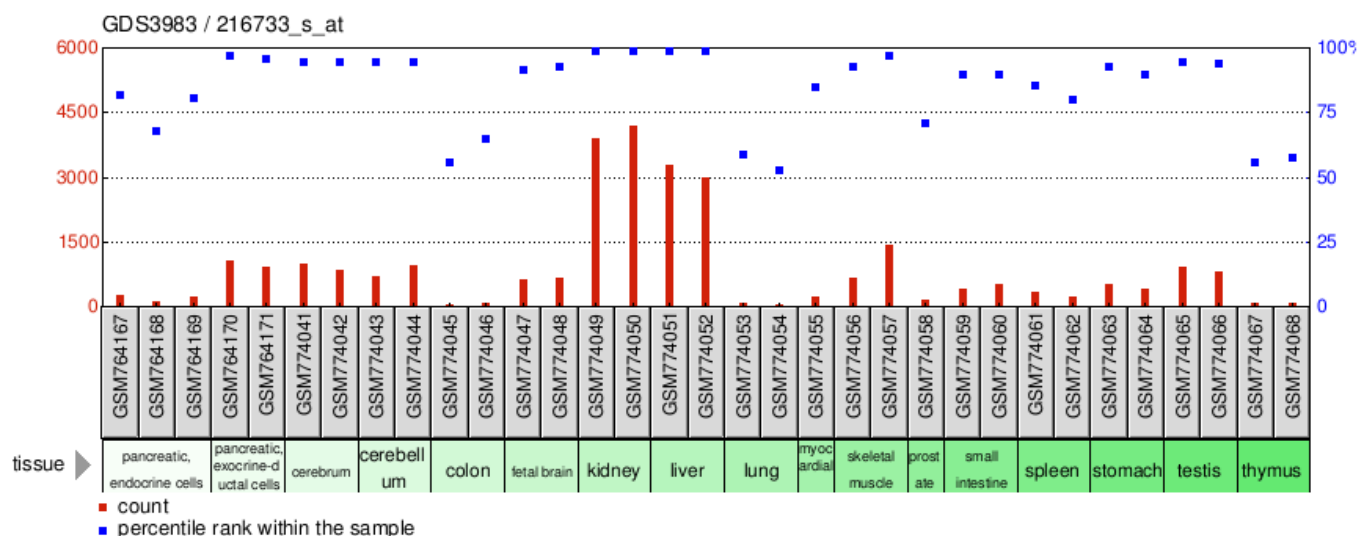
3.2. GDS724 / 36595_s_at



Interestingly, Normal-donors kidney tissues have higher expression of the gene GATM. The downregulation of this genes seems to be associated with acute transplant rejection and renal disfunction without transplant rejection.

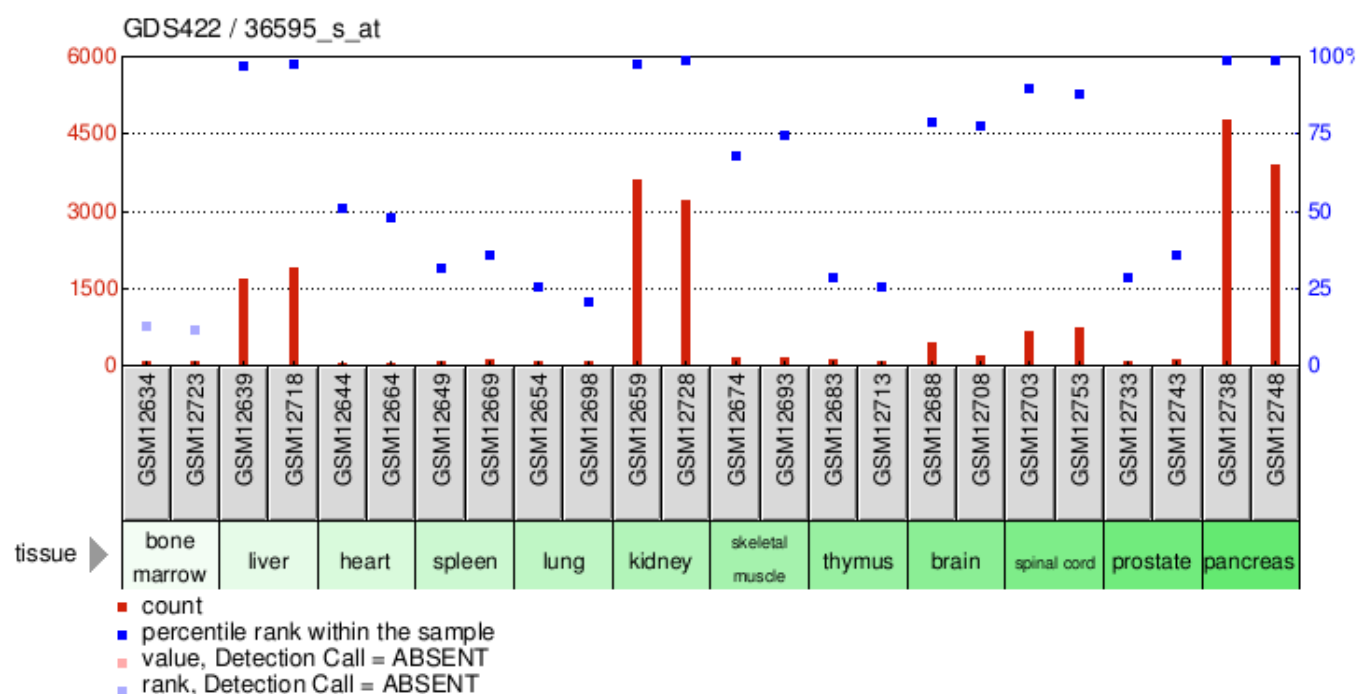
Note that the expression of the periphreal blood lymphocytes is included in the gene-profile plots above.

4. Pancreatic beta cells and panel of primary tissues - GDS3983 / 216733_s_at



The gene expression profile above confirms the higher expression of GATM gene when compared to several other human tissues.

5. Normal human tissue expression profiling (HG-U95A) - GDS422 / 36595_s_at

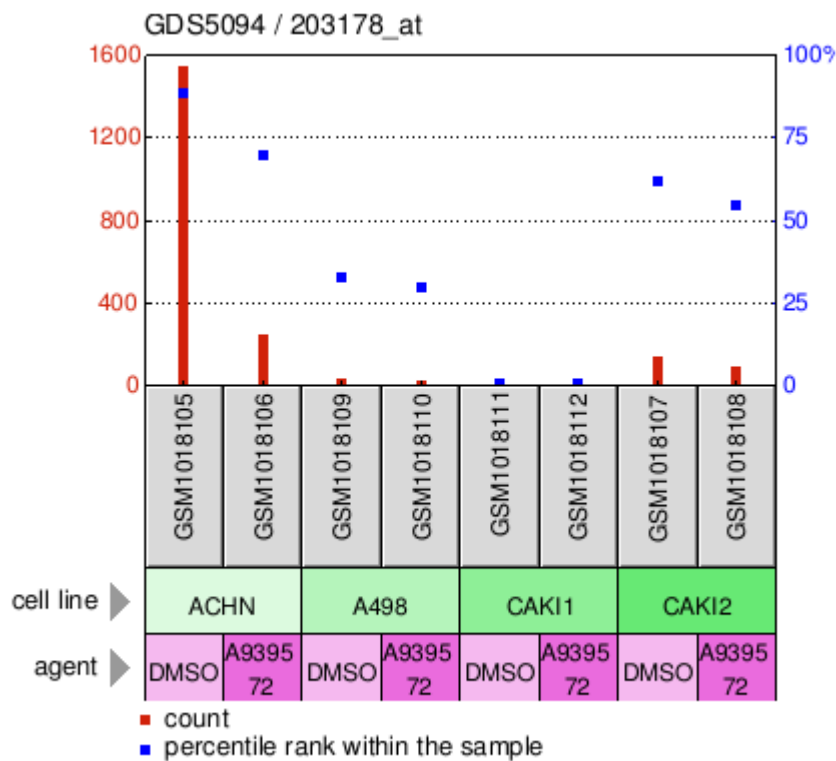


Another confirmation of the high expression of GATM in kidney and pancreas tissues.

6. Normal human tissue expression profiling (HG-U95A) - GDS181

Another two profiles for the same spot_IDs.

6.1. GDS181 / 36595_s_at

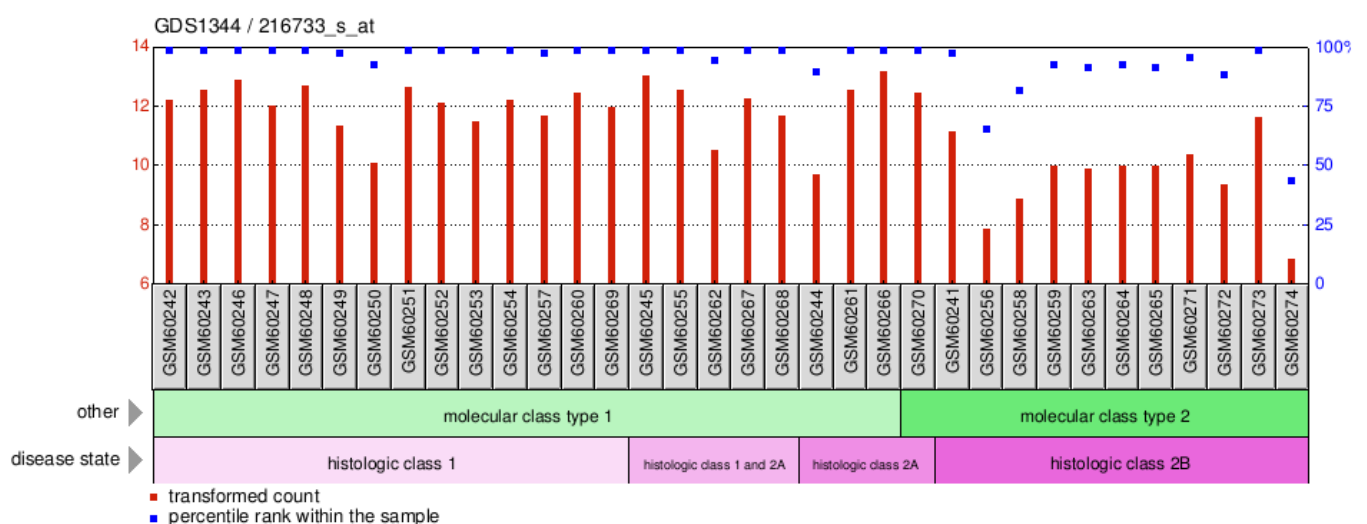


Only ACHN and CAKI2 ccRCC cells presented differential expression of the gene GATM. Interestingly, the Stearoyl-CoA desaturase 1 inhibitor treatment decreased significantly this expression, when compared to DMSO-treated ones (experiment controls).

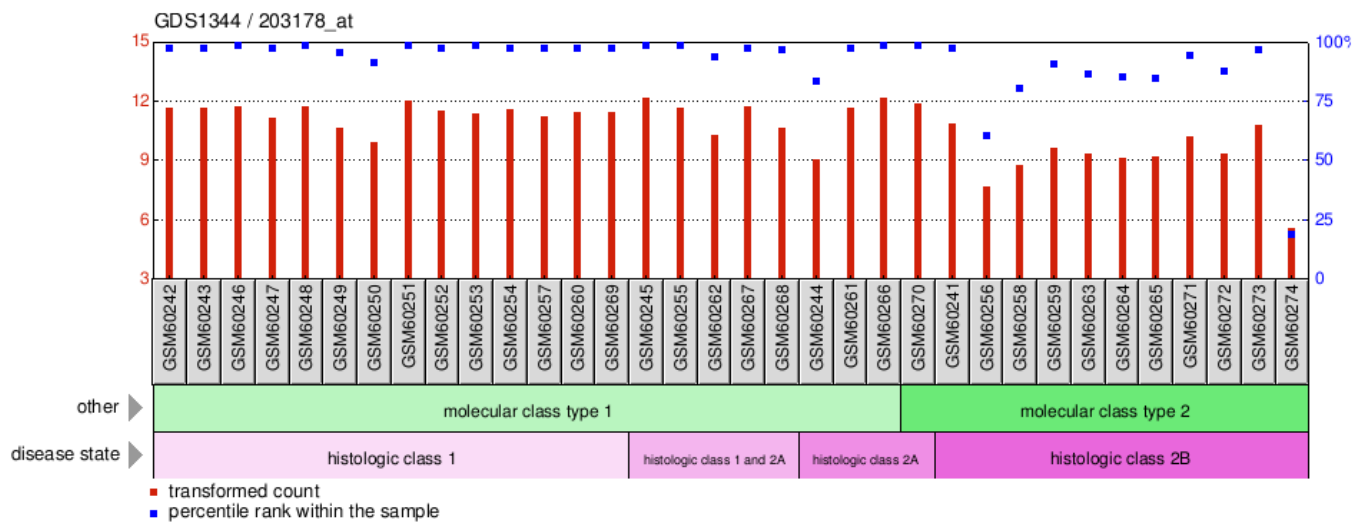
9. Papillary renal cell carcinoma classification

Two profiles for this study.

9.1. GDS1344 / 216733_s_at

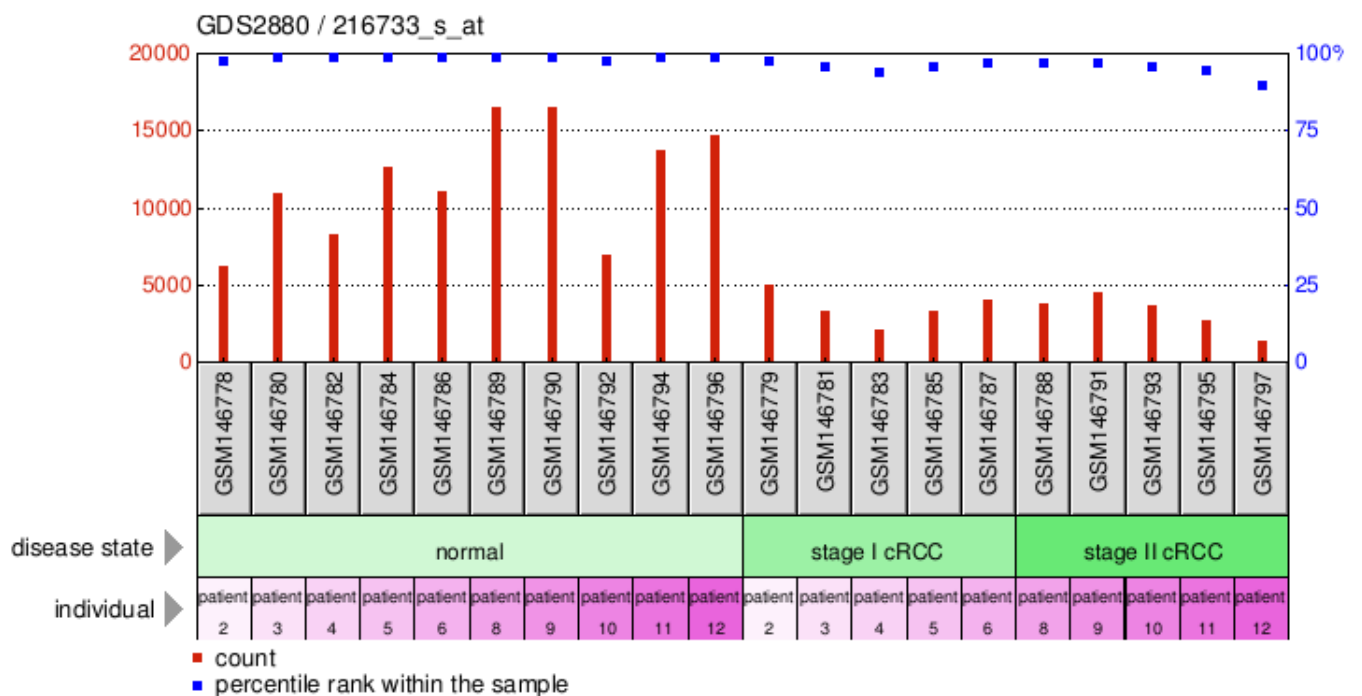


9.2. GDS1344 / 203178_at



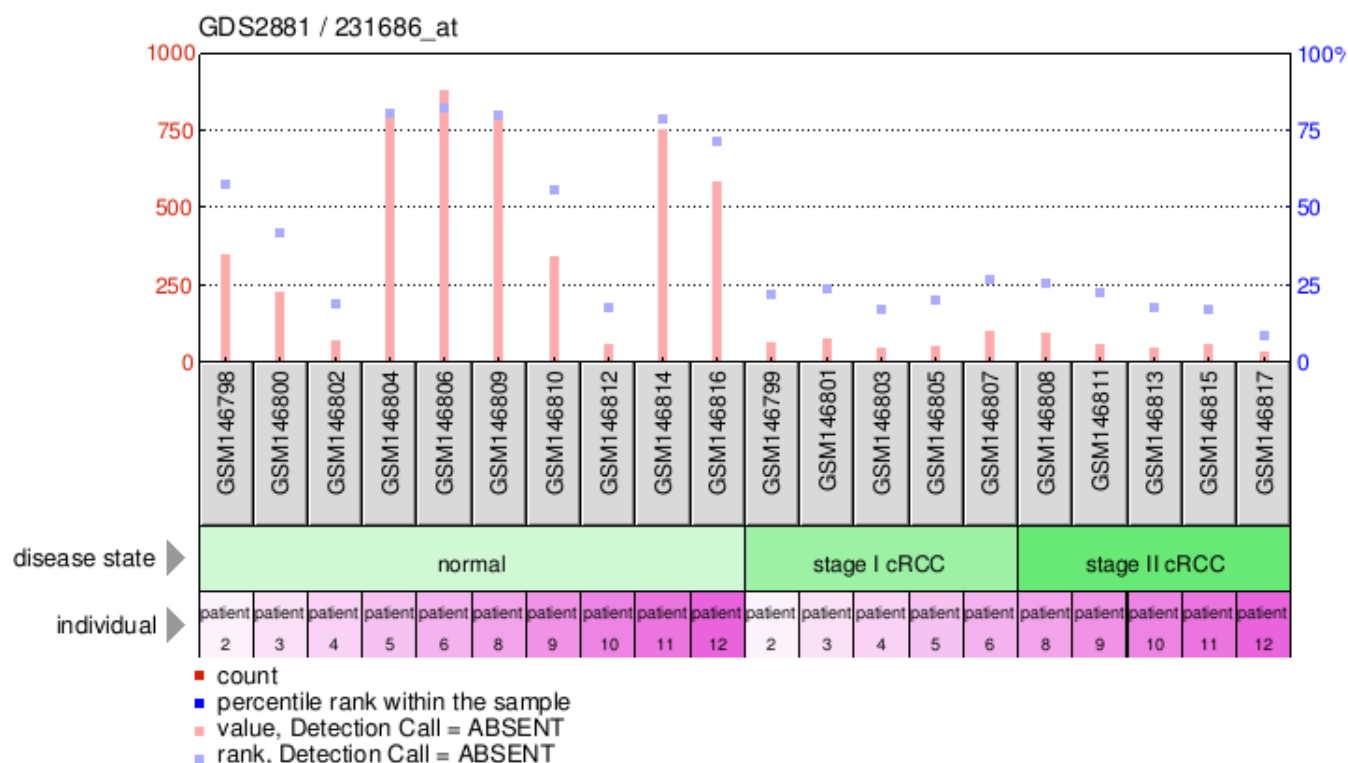
Histologic class 2B from Papillary renal cell carcinoma have a slightly downregulated expression of GATM when compared to the other types.

10. Clear cell renal cell carcinoma (HG-U133A) - GDS2880 / 216733_s_at



Normal cells have a higher expression of the GATM gene, when compared to both tested stages of cRCC.

11. Clear cell renal cell carcinoma (HB-U133B) - GDS2881 / 231686_at

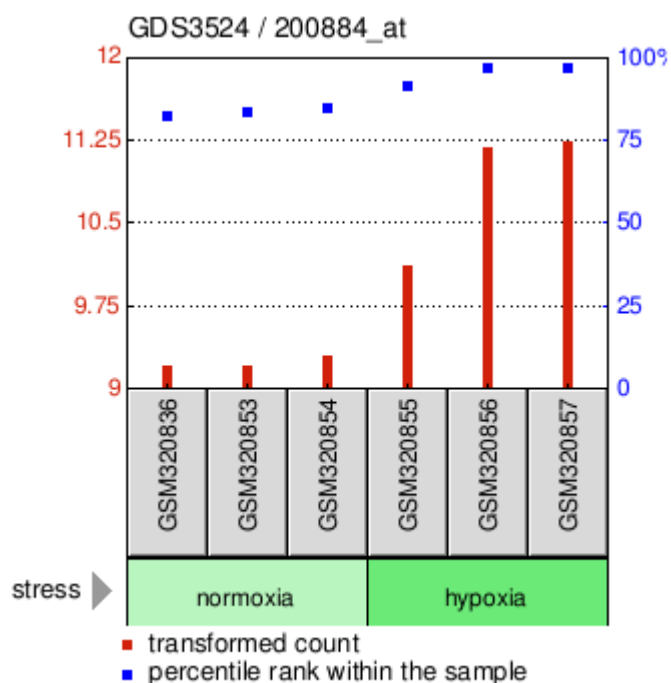


Same observation as the previous one.

CKB

With the above criteria, we retrieved just one profile for the gene CKB.

Hypoxia effect on a renal proximal tubule epithelial cell line - GDS3524 / 200884_at

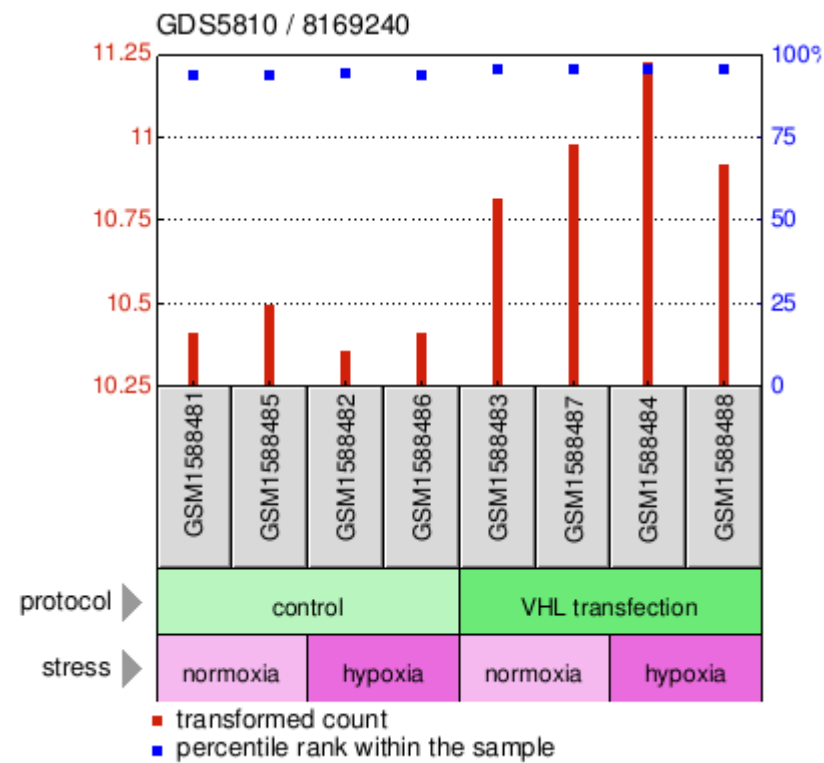


Renal proximal tubule epithelial cell lines have an increased expression of the gene CKB under hypoxia conditions, when compared to cells maintained under normoxic conditions.

PRPS1 - phosphoribosyl pyrophosphate synthetase 1

Two profiles were retrieved for the gene PRPS1, however just one from kidney samples. The other one was for whole peripheral blood samples and was not included in our study.

Hypoxia effect on von Hippel Lindau-overexpressing renal cell carcinoma cells - GDS5810 / 8169240



VHL-transfection submitted samples presented increased expression of the PRPS1 gene.

AKTs (AKT1, AKT2, and AKT3)

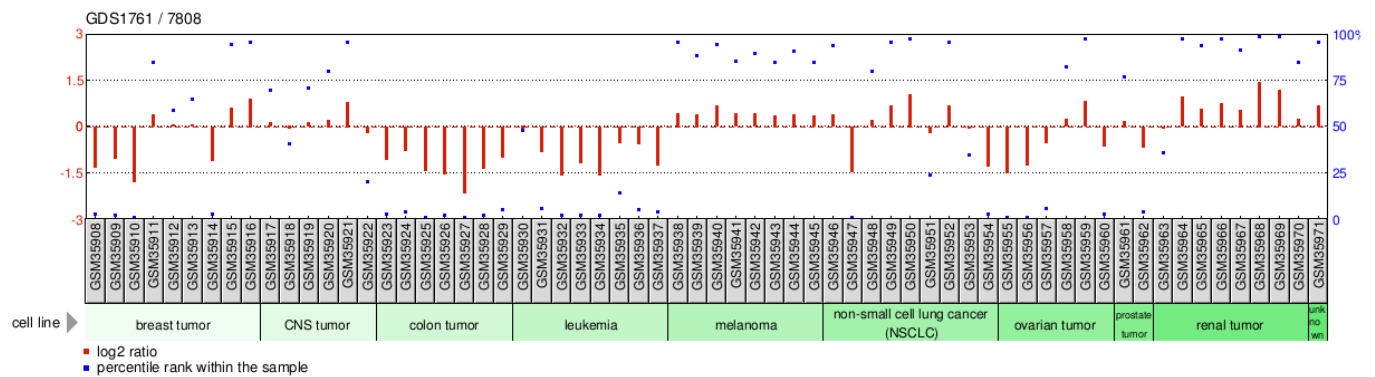
AKT1 and AKT2

No results within the search criteria.

AKT3

One result within the expression criteria.

NCI60 cancer cell lines - GDS1761 / 7808

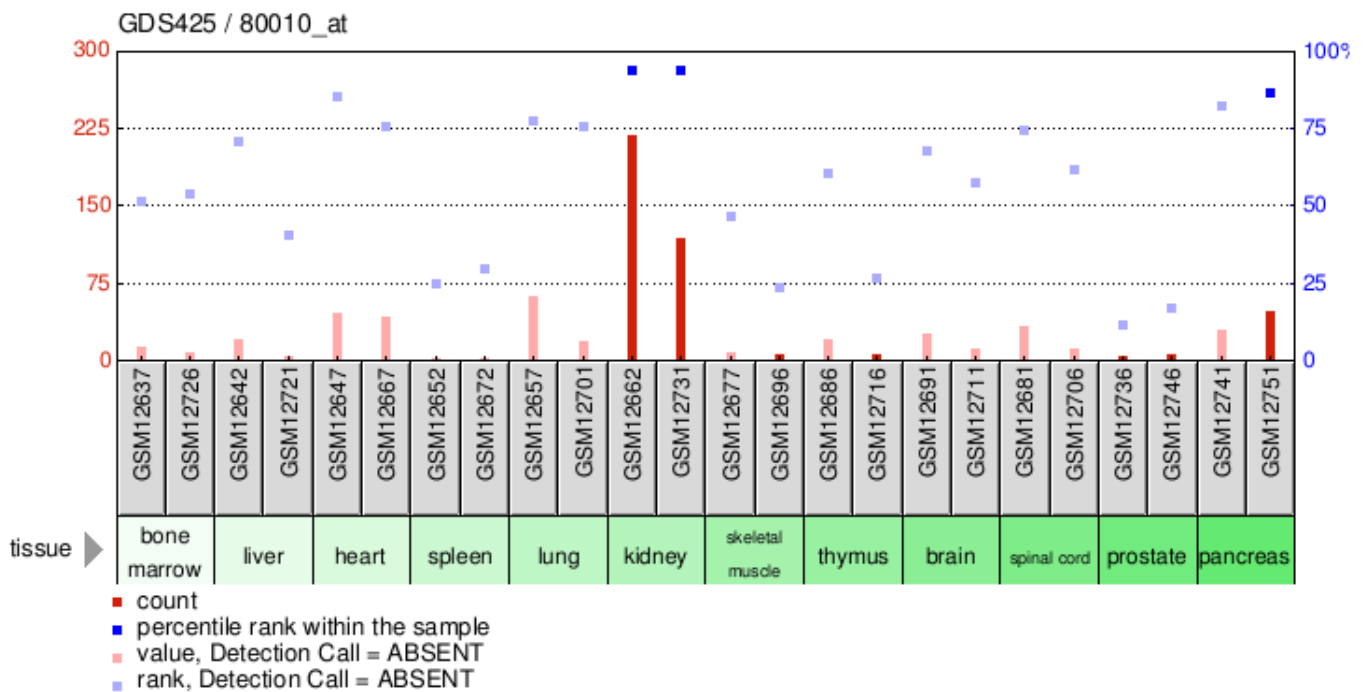


Significant upregulated expression of AKT3 on renal tumor samples.

SLC16A12 - solute carrier family 16 member 12

Just one result within the solute carrier family 6 member 8 inclusion criteria.

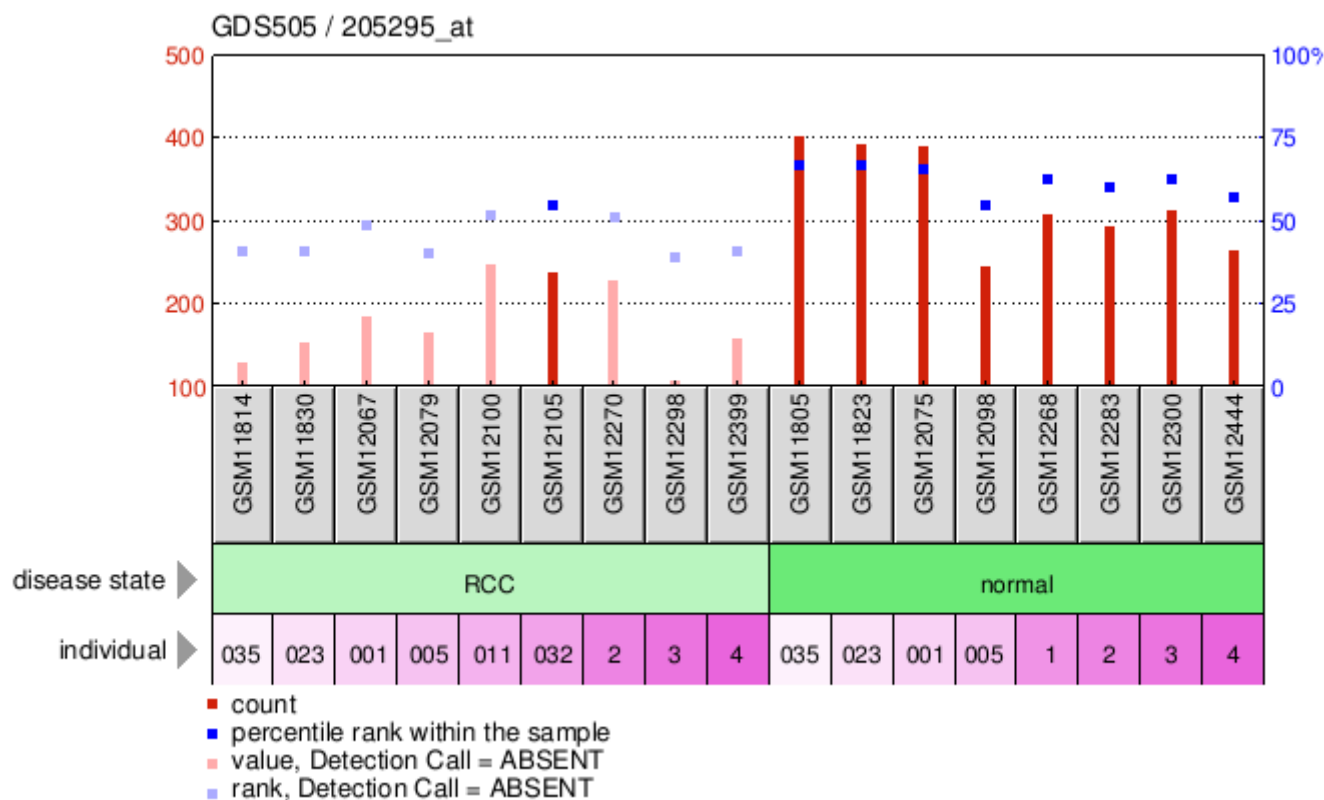
Normal human tissue expression profiling (HG-U95D) - GDS425 / 80010_at



High expression of SLC16A12 in kidney tissues.

CKMT2 - creatine kinase, mitochondrial 2

Renal clear cell carcinoma (HG-U133A) - GDS505 / 205295_at

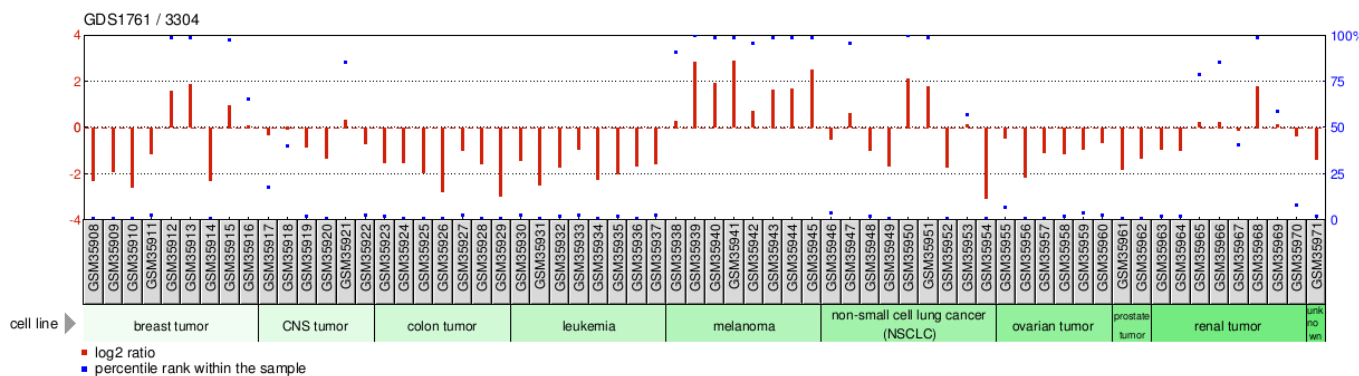


Normal cells have increased expression of the gene CKMT2 when compared to cRCC samples.

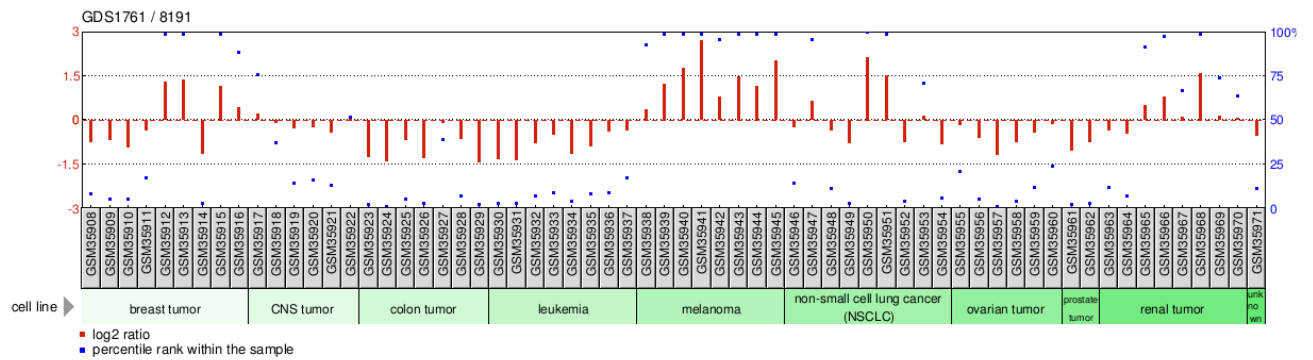
SGK1 - serum/glucocorticoid regulated kinase 1

NCI60 cancer cell lines

- GDS1761 / 3304:

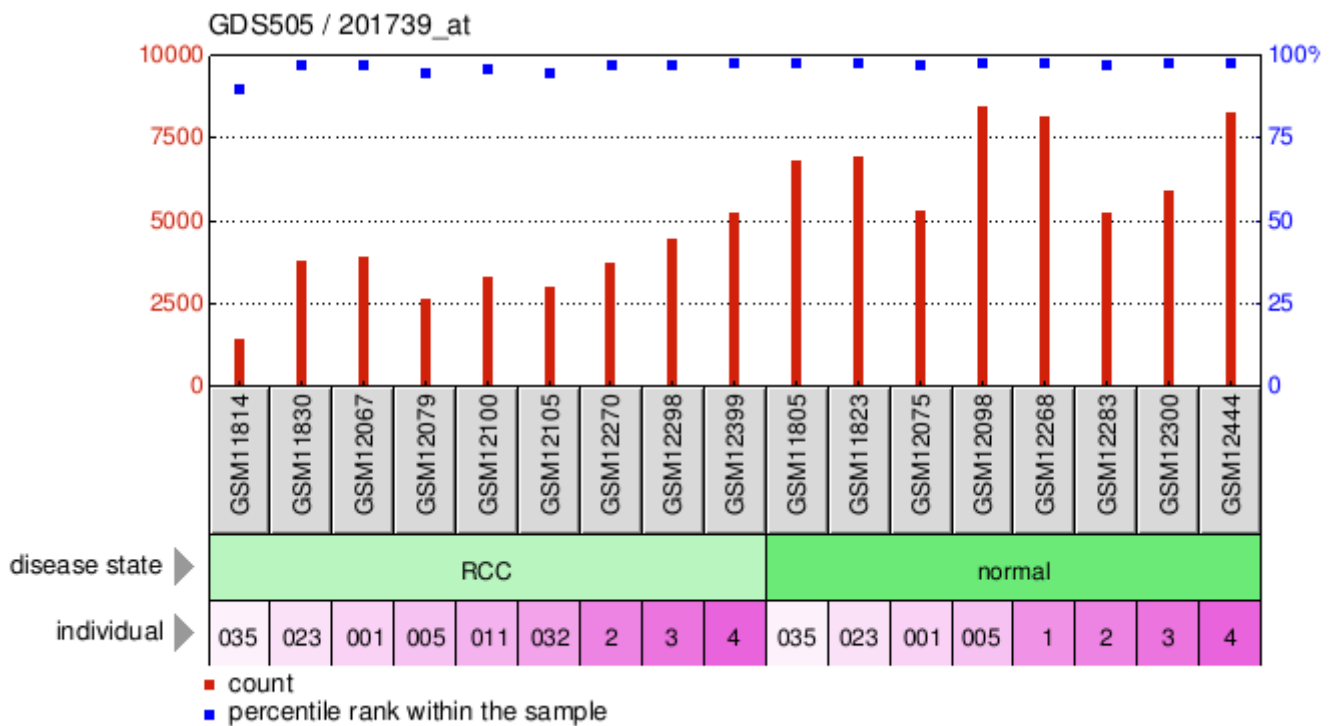


• GDS1761 / 8191:

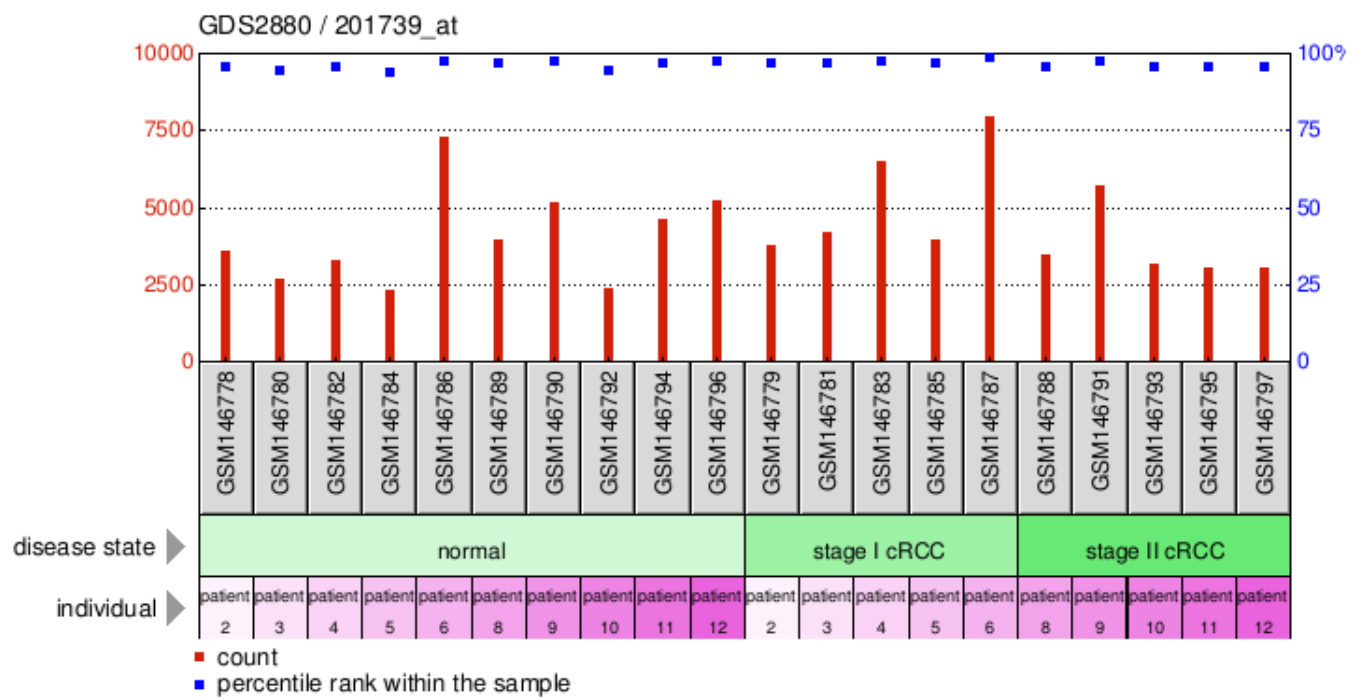


No significant differential expression from renal tumors.

Renal clear cell carcinoma (HG-U133A) - GDS505 / 201739_at



Clear cell renal cell carcinoma (HG-U133A) - GDS2880 / 201739_at



No differential expression among the samples.

PSMB5 - proteasome subunit beta 5

No results retrieved within the search criteria.

SLC6A8

No results retrieved within the search criteria. More on that later.

PSMD3

No results retrieved within the search criteria.